particle is simply “At any point in space.” If there were just one particle, the wave function would therefore assign an amplitude to every point in space. But when we have several particles, there isn’t a separate wave function for each particle. There is one big wave function, assigning a different amplitude to every possible set of locations that all the particles could be in at once. That’s how entanglement can happen; for every configuration of the particles, there is an amplitude we could square to get the probability of observing them there all at the same time.

It’s the same thing for fields, with “possible configuration of the particles” replaced by “possible configurations of the field,” where by “configuration” we now mean the values of the field at each point throughout all of space. This wave function considers every possible field configuration, and assigns an amplitude to each. If we could imagine observing the field everywhere at once, the probability of getting any particular shape of the field will be equal to the square of the amplitude assigned to that configuration.

